

The Coming Wave

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Technology, as a double-edged sword, holds immense potential to address humanity's critical challenges like climate change and aging populations but also presents unprecedented risks. Emerging fields like AI and synthetic biology introduce transformative possibilities, yet they carry tail risks that threaten societal stability. Despite extensive debates on ethics, the discourse lacks actionable containment strategies to mitigate these risks, with "pessimism aversion" often stifling serious conversations about potential negative outcomes. The challenge lies in designing governance mechanisms that balance technological progress with societal safeguards.

The exponential pace of technological evolution, from language and agriculture to modern AI, accelerates the complexities of containment. The integration of powerful technologies into everyday life amplifies their potential for harm, especially as creators lose control over their unintended consequences. This proliferation underscores the need for a robust "containment architecture" involving technical, legal, cultural, and political measures. Without such frameworks, technologies risk destabilizing societies by empowering malicious actors and exposing vulnerabilities in interconnected global systems.

Historical precedents, such as nuclear containment, demonstrate that effective governance is possible but rare. The wave of AI and synthetic biology, however, is far more pervasive and rapid, demanding innovative approaches to governance. Recursive technological advancements, where AI designs better AI or synthetic biology iterates on life, create self-perpetuating cycles that evade traditional control mechanisms. The democratization of these capabilities further blurs the lines between innovation and risk, as access to such tools becomes increasingly widespread.

The current geopolitical and economic incentives exacerbate the challenge of containment. Nation-states compete for technological supremacy, often prioritizing innovation over regulation. Simultaneously, private corporations drive technological development through massive R&D investments, yet their motivations align more with profit than societal well-being. These dynamics foster a fragmented regulatory environment ill-equipped to handle the cross-disciplinary and international implications of frontier technologies.

AI, synthetic biology, and automation are transforming industries and labor markets, presenting both opportunities and disruptions. These tools, while initially augmenting human capabilities, are fundamentally labor-replacing, threatening jobs in cognitive and manual domains. Furthermore, they amplify societal fragility by embedding risks at every level—economic, social, and political—while undermining the traditional nation-state's ability to maintain order and cohesion.

Containment requires a multifaceted strategy, including investments in technical safety akin to an "Apollo program" for AI and biosafety, rigorous audits to ensure transparency, and the establishment of regulatory choke points to decelerate development. Critics must transition from abstract critiques to practical solutions, while businesses should align profit motives with ethical governance. Governments need to reclaim leadership in technological innovation, regulating with foresight and engaging in public-private partnerships to manage risks.

International collaboration, modeled on successful treaties like the Montreal Protocol, is essential for global containment. Alliances between nations, particularly rivals like the U.S. and China, can harmonize standards and create joint oversight mechanisms. However, this requires overcoming political and economic barriers to prioritize collective safety over national or corporate interests. Movements advocating for responsible technological development can provide the necessary public pressure to drive these initiatives.

The cultural mindset surrounding failure and experimentation must shift to encourage transparency and learning without fear of backlash. Containment also depends on fostering a global consensus around ethical boundaries, such as limiting autonomy and recursive self-improvement in AI. Open-source philosophies, while valuable in many contexts, should be restricted for high-stakes technologies to prevent uncontrolled proliferation.

Ultimately, technology should be viewed as a means to enrich human life rather than an end in itself. Its integration into society demands continuous oversight, adaptation, and a commitment to balance its transformative power with the values and structures that underpin civilization. As the authors argue, the future of humanity hinges on navigating the delicate path between technological catastrophe and a dystopian surrender to unchecked innovation.

Excerpts

Location: 464

What if we could distill the essence of what makes us humans so productive and capable into software, into an algorithm? Finding the answer might unlock unimaginably powerful tools to help tackle our most intractable problems. Here might be a tool, an impossible but extraordinary tool, to help us get through the awesome challenges of the decades ahead, from climate change to aging populations to sustainable food.

Location: 500

AI, synthetic biology, and other advanced forms of technology produce tail risks on a deeply concerning scale.

Location: 520

The current discourse around technology ethics and safety is inadequate. Despite the many books, debates, blog posts, and tweetstorms about technology, you rarely hear anything about containing it. I see this as an interlocking set of technical, social, and legal mechanisms constraining and controlling technology working at every possible level: a means, in theory, of evading the dilemma. Yet even technology's harshest critics tend to dodge this language of hard containment.

Location: 561

Why do some get snarky and accuse people who raise these questions of catastrophizing or of "overlooking the amazing good" of technology? This widespread emotional reaction I was observing is something I have come to call the pessimism-aversion trap: the misguided analysis that arises when you are overwhelmed by a fear of confronting potentially dark realities, and the resulting tendency to look the other way.

Location: 566

Our species is not wired to truly grapple with transformation at this scale, let alone the potential that technology might fail us in this way.

Location: 587

Pessimism aversion is an emotional response, an ingrained gut refusal to accept the possibility of seriously destabilizing outcomes. It tends to come from those in secure and powerful positions with entrenched worldviews, people who can superficially cope with change but struggle to accept any real challenge to their world order. Many of those whom I accuse of being stuck in the pessimism-aversion trap fully embrace the growing critiques of technology. But they nod along without actually taking any action. We'll manage, we always do, they say.

Location: 610

But without containment, every other aspect of technology, every discussion of its ethical shortcomings, or the benefits it could bring, is inconsequential. We urgently need watertight answers for how the coming wave can be controlled and contained, how the safeguards and affordances of the democratic nation-state can be maintained, but right now no one has such a plan.

Location: 638

If technology damages human lives, or produces societies filled with harm, or renders them ungovernable because we empower a chaotic long tail of bad (or unintentionally dangerous) actors—if, in the aggregate, technology is damaging—then it can be said to have failed in another, deeper sense, failing to live up to its promise. Failure in this sense isn't intrinsic to technology; it is about the context within which it operates, the governance structures it is subject to, the networks of power and uses to which it is put.

Location: 733

The irony of general-purpose technologies is that, before long, they become invisible and we take them for granted. Language, agriculture, writing¹²—each was a general-purpose technology at the center of an early wave. These three waves formed the foundation of civilization as we know it. Now we take them for granted.

Location: 767

The ten thousand years¹⁸ up to 1000 BCE saw seven general-purpose technologies emerge. The two hundred years between 1700 and 1900 marked the arrival of six, from steam engines to electricity. And in the last hundred¹⁹ years alone there were seven.

Location: 887

Technology's unavoidable challenge is that its makers quickly lose control over the path their inventions take once introduced to the world.

Location: 910

The more powerful a technology, the more ingrained it is in every facet of life and society. Thus, technology's problems have a tendency to escalate in parallel with its capabilities, and so the need for containment grows more acute over time.

Location: 930

Think of containment, then, as a set of interlinked and mutually reinforcing technical, cultural, legal, and political mechanisms for maintaining societal control of technology during a time of exponential change; an architecture up to the task of containing what would have once been centuries or millennia of technological change happening now in a matter of years or even months, where consequences ricochet around the world in seconds.

Location: 972

The Luddites were no more successful at stopping new industrial technologies than horse owners and carriage makers were at preventing cars. Where there is demand, technology always breaks out, finds traction, builds users.

Location: 1,066

Nuclear is an exception to the unstoppable spread of technology, but only because of the tremendous costs and complexity involved, the decades of tough multilateral effort, the fear-inducing enormity of its lethal potential, and pure luck.

Location: 1,072

Glimmers of containment are rare and often flawed. They include moratoriums on biological and chemical weapons; the Montreal Protocol of 1987, which phased out substances damaging the atmosphere's ozone layer, particularly CFCs; the EU's ban on genetically modified organisms in foodstuffs; and a self-organized moratorium on human gene editing.

Location: 1,191

First bits and then increasingly genes supplanted atoms as the building blocks of invention.

Location: 1,196

The coming wave of technology is built primarily on two general-purpose technologies capable of operating at the grandest and most granular levels alike: artificial intelligence and synthetic biology. For the first time core components of our technological ecosystem directly address two foundational properties of our world: intelligence and life.

Location: 1,207

The coming wave will be more difficult to contain than any in history, more fundamental, more far-reaching. Understanding the wave and its contours is critical to assessing what awaits us in the twenty-first century.

Location: 1,211

The more technologies there are,³ the more they can in turn become components of other new technologies so that, in the words of the economist W. Brian Arthur, “the overall collection of technologies bootstraps itself upward from the few to the many and from the simple to the complex.” Technology is hence like a language or chemistry: not a set of independent entities and practices, but a commingling set of parts to combine and recombine.

Location: 1,221

Another trait of the new wave is speed. The engineer and futurist Ray Kurzweil⁵ talks about the “law of accelerating returns,” feedback loops where advances in technology further increase the pace of development. By allowing work at greater levels of complexity and precision, more sophisticated chips and lasers help create more powerful chips, for example, which in turn can produce better tools for further chips. We see this now on⁶ a large scale, with AI helping design better chips and production techniques that enable more sophisticated forms of AI and so on. Different parts of the wave spark and accelerate one another, sometimes with extreme unpredictability and combustibility.

Location: 1,295

Within a few years AIs will be able to talk about, reason over, and even act in the same world that we do. Their sensory systems will be as good as ours. This does not equate to superintelligence (more on that below), but it does make for incredibly powerful systems. It means that AI will become inextricably part of the social fabric.

Location: 1,366

a key ingredient of the LLM revolution is that for the first time very large models could be trained directly on raw, messy, real-world data, without the need for carefully curated and human-labeled data sets.

Location: 1,373

More realistically, the average American¹⁷ reads a book for about fifteen minutes per day, which over the year amounts to reading about a million words. That’s roughly six orders of magnitude less than what these models consume in a single monthlong training run.

Location: 1,386

Over the last decade the amount of computation used to train the largest models has increased exponentially. Google’s PaLM uses²⁰ so much that were you to have a drop of water for every floating-point operation (FLOP) it used during training, it would fill the Pacific.

Location: 1,390

in less than ten years the amount of compute used to train the best AI models has increased by nine orders of magnitude—going from two petaFLOPs to ten billion petaFLOPs. To get a sense of one petaFLOP, imagine a billion people each holding a million calculators, doing a complex multiplication, and hitting “equals” at the same time.

Location: 1,437

nimble creators working with efficient systems, curated data sets, and quick iterations can already quickly rival the most well-resourced developers.

Location: 1,555

Today, as the LaMDA-sentience saga illustrates, systems are already close to passing the Turing test. But, as many have pointed out, intelligence is about so much more than just language (or indeed any other single facet of intelligence taken in isolation). One particularly important dimension is in the ability to take actions. We don't just care about what a machine can say; we also care about what it can do. What we would really like to know is, can I give an AI an ambiguous, open-ended, complex goal that requires interpretation, judgment, creativity, decision-making, and acting across multiple domains, over an extended time period, and then see the AI accomplish that goal? Put simply, passing a Modern Turing Test would involve something like the following: an AI being able to successfully act on the instruction "Go make \$1 million on Amazon in a few months with just a \$100,000 investment." It might research the web to look at what's trending, finding what's hot and what's not on Amazon Marketplace; generate a range of images and blueprints of possible products; send them to a drop-ship manufacturer it found on Alibaba; email back and forth to refine the requirements and agree on the contract; design a seller's listing; and continually update marketing materials and product designs based on buyer feedback. Aside from the legal requirements of registering as a business on the marketplace and getting a bank account, all of this seems to me eminently doable. I think it will be done³⁵ with a few minor human interventions within the next year, and probably fully autonomously within three to five years.

Location: 1,651

Thanks to ever-improving techniques,⁵ the cost of human genome sequencing fell from \$1 billion in 2003 to well under \$1,000 by 2022. That is, the price dropped⁶ a millionfold in under twenty years, a thousand times faster than Moore's law.

Location: 1,679

Today technologies like CRISPR¹³ are simple and cheap to use; they have, in the words of the biologist Nessa Carey, "democratized biological science." Experiments that once took years are tackled by grad students in weeks. Companies like the Odin will sell you a genetic engineering kit including live frogs and crickets for \$1,999, while another kit includes a mini-centrifuge, a polymerase chain reaction machine, and all the reagents and materials you need to get going.

Location: 1,685

You can now buy a benchtop¹⁴ DNA synthesizer (see the next section) for as little as \$25,000 and use it as you wish, without restriction or oversight, at home in your bio-garage.

Location: 1,700

A world of possibility for the creation of DNA has opened up, one in which cycles of designing, building, testing, and iterating happen at a radically accelerated pace.

Location: 1,708

In the words of the Stanford bioengineer²⁰ Drew Endy, “Biology is the ultimate distributed manufacturing platform.”

Location: 1,768

Scientists like the Nobel laureate Frances Arnold create enzymes that produce novel chemical reactions, including ways to bind silicon and carbon, usually a tricky, energy-intensive process with wide-ranging uses in areas like electronics. Arnold’s method is fifteen times³³ as energy efficient as standard industrial alternatives. The next step involves scaling up production of biological materials and processes. In this way significant products like meat replacements or new materials sucking carbon out of the atmosphere could be grown as much as made.

Location: 1,772

The vast petrochemical industry could see a challenge from young start-ups like Solugen, whose Bioforge is an attempt to build a carbon-negative factory; it would produce a wide range of chemicals and commodities, from cleaning products to food additives to concrete, all while pulling carbon out of the atmosphere. Their process is essentially low-energy, low-waste bio-manufacturing at industrial scale, built on AI and biotech. Another company, LanzaTech, harnesses genetically modified bacteria to convert waste CO₂ from steel mill production into widely used industrial chemicals.

Location: 1,785

Theoretically, the entirety of the world’s³⁶ data might be stored in just one kilogram of DNA. A biological version of a transistor called a transcriptor uses DNA and RNA molecules to act as logic gates. There is still a long way to go before this technology can be harnessed. But all the functional parts of a computer—data storage, information transmission, and a basic system of logic—can in principle be replicated using biological materials.

Location: 1,890

From drones watching livestock to precision irrigation rigs to small mobile robots patrolling vast indoor farms, from seeding to harvesting, picking to palletizing, watering tomatoes to tracking and herding cattle, the reality of the food we eat today is that it increasingly comes from a world of robots, driven by AI, currently being rolled out and scaled up.

Location: 2,036

AT ITS CORE, THE coming wave is a story of the proliferation of power. If the last wave reduced the costs of broadcasting information, this one reduces the costs of acting on it, giving rise to technologies that go from sequencing to synthesis, reading to writing, editing to creating, imitating conversations to leading them. In this, it is qualitatively different from every previous wave, despite all the big claims made about the transformative power of the internet. This kind of power is even harder to centralize and oversee; this wave is not just a deepening and acceleration of history’s pattern, then, but also a sharp break from it.

Location: 2,128

Just as globalized and highly connected markets transmit contagion in a financial crisis, so with technology. Network scale makes containing damage, if or when it comes, almost impossible. Interlinked global systems are containment nightmares. And we already live in an age of interlinked global systems. In the coming wave a single point—a given program, a genetic change—can alter everything.

Location: 2,291

Nobody really knows how we can contain the very features being researched so intently in the coming wave. There comes a point where technology can fully direct its own evolution; where it is subject to recursive processes of improvement; where it passes beyond explanation; where it is consequently impossible to predict how it will behave in the wild; where, in short, we reach the limits of human agency and control.

Location: 2,339

Technological rivalry is a geopolitical reality. Indeed it always has been. Nations feel the existential need to keep up with their peers. Innovation is power. Second comes a global research ecosystem with its ingrained rituals rewarding open publication, curiosity, and the pursuit of new ideas at all costs. Then come the immense financial gains from technology and the urgent need to tackle our global social challenges. And the final driver is perhaps the most human of all: ego.

Location: 2,438

For a long time I objected, resisting the framing of technological progress as a zero-sum international arms race. At DeepMind, I always pushed back on references to us as a Manhattan Project for AI, not just because of the nuclear comparison, but because even the framing might initiate a series of other Manhattan Projects, feeding an arms race dynamic when close global coordination, break points, and slowdowns were needed. But the reality is that the logic of nation-states is at times painfully simple and yet utterly inevitable. In the context of a state's national security, merely floating an idea becomes dangerous. Once the words are out, the starting gun is fired, the rhetoric itself producing a drastic national response. And then it spirals.

Location: 2,527

Worldwide R&D spending³³ is at well over \$700 billion annually, hitting record highs. Amazon's R&D budget alone³⁴ is \$78 billion, which would be the ninth biggest in the world if it were a country. Alphabet, Apple, Huawei, Meta,³⁵ and Microsoft all spend well in excess of \$20 billion a year on R&D.

Location: 2,534

Both the imperative of openness and the sheer mass of easily available research material mean this is an inherently deep-rooted and widely distributed set of incentives and foundations for future research that no one can fully govern.

Location: 2,586

Carlota Perez sees an equivalent⁴² “frenzy phase” as being part of every major technology rollout for at least the last two hundred years, from the original telephone cables to contemporary high-bandwidth internet. The boom never lasts, but the raw speculative drive produces lasting change, a new technological substrate.

Location: 2,591

Put simply: most technology is made to earn money. If anything, this is perhaps the most persistent, entrenched, dispersed incentive of all.

Location: 2,620

The coming wave represents the greatest economic prize in history. It is a consumer cornucopia and potential profit center without parallel. Anyone looking to contain it must explain how a distributed, global, capitalist system of unbridled power can be persuaded to temper its acceleration, let alone leave it on the table.

Location: 2,706

The energy scholar Vaclav Smil calls ammonia, cement, plastics, and steel the four pillars of modern civilization: the material base underwriting modern society, each hugely carbon-intensive to produce, with no obvious successors. Without these materials modern life stops, and without fossil fuels the materials stop.

Location: 2,735

A school of naive techno-solutionism sees technology as the answer to all of the world’s problems. Alone, it’s not. How it is created, used, owned, and managed all make a difference. No one should pretend that technology is a near-magical answer to something as multifaceted and immense as climate change. But the idea that we can meet the century’s defining challenges without new technologies is completely fanciful.

Location: 2,782

Technology is now an indispensable mega-system infusing every aspect of daily life, society, and the economy. No one can do without it. Entrenched incentives are in place for more of it, radically more. No one is in full control of what it does or where it goes next.

Location: 2,815

Given that nation-states are charged with managing and regulating the impact of technology in the best interests of their populations, how prepared are they for what’s to come? If the state is unable to coordinate the containment of this wave, unable to ensure it is of net benefit to its citizens, what options does that leave humanity in the medium to long term?

Location: 2,820

The political order that fostered rising wealth, better living standards, growing education, science, and technology, a world tending toward peace, is now under immense strain, destabilized in part by the very forces it helped engender.

Location: 2,918

Behind the new authoritarian impulse and political instability lies a growing pool of social resentment. A key catalyst of instability¹¹ and social resentment, inequality has surged across Western nations in recent decades, and nowhere more so than in the United States. Between 1980 and 2021¹² the share of national income earned by the top 1 percent has almost doubled and now sits just under 50 percent.

Location: 2,952

A meta-analysis published¹⁸ in the journal *Nature* reviewed the results of nearly five hundred studies, concluding there is a clear correlation between growing use of digital media and rising distrust in politics, populist movements, hate, and polarization. Correlation may not be causation, but this systematic review throws up “clear evidence of serious threats to democracy” coming from new technologies.

Location: 2,959

Too fast in its development, too global, too protean and enticing for any simple model of containment, strategically critical, relied upon by billions, modern technology itself is a prime actor, a monumental force nation-states struggle to manage. AI, synthetic biology, and the rest are being introduced to dysfunctional societies already rocked back and forth on technological waves of immense power.

Location: 2,966

As the historian of technology¹⁹ Langdon Winner puts it, “Technology in its various manifestations is a significant part of the human world. Its structures, processes, and alterations enter into and become part of the structures, processes, and alterations of human consciousness, society, and politics.” In other words, technology is political.

Location: 3,011

What emerges will, I think, tend in two directions with a spectrum of outcomes in between. On one trajectory, some liberal²³ democratic states will continue to be eroded from within, becoming a kind of zombie government. Trappings of liberal democracy and the traditional nation-state remain, but functionally they are hollowed out, the core services increasingly threadbare, the polity unstable and fractious. Lurching on in the absence of anything else, they become ever more degraded and dysfunctional. On another, unthinking adoption of some aspects of the coming wave opens pathways to domineering state control, creating supercharged Leviathans whose power goes beyond even history’s most extreme totalitarian governments. Authoritarian regimes may also tend toward zombie status, but equally they may double down, get boosted, become fully fledged techno-dictatorships. On either path, the delicate balance holding states together is tipped into chaos. Both failing states and authoritarian regimes are disastrous outcomes, not just on their own terms, but also for governing technology; neither flailing bureaucracies, populist opportunists, nor all-powerful dictators are people you’d want to be fundamentally responsible for controlling powerful new technologies. Neither direction can or will contain the coming wave.

Location: 3,066

Here is a parable for technology in the twenty-first century. Software created by the security services of the world's most technologically sophisticated state is leaked or stolen. From there it finds its way into the hands of digital terrorists working for one of the world's most failed states and capricious nuclear powers. It is then weaponized, turned against the core fabric of the contemporary state: health services, transport and power infrastructures, essential businesses in global communications and logistics. In other words, thanks to a basic failure of containment, a global superpower became a victim of its own powerful and supposedly secure technology. This is uncontained asymmetry in action.

Location: 3,079

Now imagine if, instead of accidentally leaving open a loophole, the hackers behind WannaCry had designed the program to systematically learn about its own vulnerabilities and repeatedly patch them. Imagine if, as it attacked, the program evolved to exploit further weaknesses. Imagine that it then started moving through every hospital, every office, every home, constantly mutating, learning. It could hit life-support systems, military infrastructure, transport signaling, the energy grid, financial databases. As it spread, imagine the program learning to detect and stop further attempts to shut it down. A weapon like this is on the horizon if not already in development.

Location: 3,094

Technology is ultimately political because technology is a form of power. And perhaps the single overriding characteristic of the coming wave is that it will democratize access to power.

Location: 3,096

I think of it like this: just as the costs of processing and broadcasting information plummeted in the consumer internet era, the cost of actually doing something, taking action, projecting power, will plummet with the next wave.

Location: 3,106

Today, no matter how wealthy you are, you simply cannot buy a more powerful smartphone than is available to billions of people. This phenomenal achievement of civilization is too often overlooked. In the next decade, access to ACIs will follow the same trend. Those same billions will soon have broadly equal access to the best lawyer, doctor, strategist, designer, coach, executive assistant, negotiator, and so on. Everyone will have a world-class team on their side and in their corner.

Location: 3,111

If everyone has access to more capability, that clearly also includes those who wish to cause harm. With technology evolving faster than defensive measures, bad actors, from Mexican drug cartels to North Korean hackers, are given a shot in the arm. Democratizing access necessarily means democratizing risk.

Location: 3,138

THE COST OF MILITARY-GRADE drones⁸ has fallen by three orders of magnitude over the last decade. By 2028, \$26 billion a year⁹ will be spent on military drones, and at that point many are likely to be fully autonomous.

Location: 3,152

Just as systems like AlphaGo learn unexpected strategies from millions of self-played games, so too will AI-enabled cyberattacks. However much you war-game every eventuality, there's inevitably going to be a tiny vulnerability discoverable by a persistent AI.

Location: 3,157

As the cybersecurity expert¹² Bruce Schneier has pointed out, AIs could digest the world's laws and regulations to find exploits, arbitrating legalities. Imagine a huge cache of documents from a company leaked. A legal AI might be able to parse this against multiple legal systems, figure out every possible infraction, and then hit that company with multiple crippling lawsuits around the world at the same time.

Location: 3,162

AI adept at exploiting not just financial, legal, or communications systems but also human psychology, our weaknesses and biases, is on the way. Researchers at Meta created¹³ a program called CICERO. It became an expert at playing the complex board game Diplomacy, a game in which planning long, complex strategies built around deception and backstabbing is integral. It shows how AIs could help us plan and collaborate, but also hints at how they could develop psychological tricks to gain trust and influence, reading and manipulating our emotions and behaviors with a frightening level of depth, a skill useful in, say, winning at Diplomacy or electioneering and building a political movement.

Location: 3,173

Just as social media's one-to-many broadcast effect means a single person can suddenly broadcast globally, so the capacity for far-reaching consequential action is becoming available to everyone. This new dynamic—where bad actors are emboldened to go on the offensive—opens up new vectors of attack thanks to the interlinked, vulnerable nature of modern systems: not just a single hospital but an entire health system can be hit; not just a warehouse but an entire supply chain. With lethal autonomous weapons the costs, in both material and above all human terms, of going to war, of attacking, are lower than ever. At the same time, all this introduces greater levels of deniability and ambiguity, degrading the logic of deterrence.

Location: 3,185

If we feel that the fundamentals—the electricity running our houses, the transport systems getting us around, the energy networks keeping us warm, our personal everyday security—are falling to pieces and there is nothing either we or the government can do, a foundation of the system is chipped away. If the state began in new forms of war, perhaps it will end in the same way.

Location: 3,194

Unlike an arrow or even a hypersonic missile, AI and bioagents will evolve more cheaply, more rapidly, and more autonomously than any technology we've ever seen. Consequently, without a dramatic set of interventions to alter the current course, millions will have access to these capabilities in just a few years.

Location: 3,240

Eventually entire and rich synthetic histories of seemingly real-world events will be easy to generate. Individual citizens won't have time or the tools to verify a fraction of the content coming their way. Fakes will easily pass sophisticated checks, let alone a two-second smell test.

Location: 3,253

Early in the COVID-19 pandemic a blizzard of disinformation had deadly consequences. A Carnegie Mellon study analyzed more than 200 million tweets discussing COVID-19 at the height of the first lockdown. Eighty-two percent of influential users²⁷ advocating for "reopening America" were bots. This was a targeted "propaganda machine," most likely Russian, designed to intensify the worst public health crisis in a century.

Location: 3,262

The rise of synthetic media at scale and minimal cost amplifies both disinformation (malicious and intentionally misleading information) and misinformation (a wider and more unintentional pollution of the information space) at once. Cue an "Infocalypse,"²⁸ the point at which society can no longer manage a torrent of sketchy material, where the information ecosystem grounding knowledge, trust, and social cohesion, the glue holding society together, falls apart. In the words of a Brookings Institution²⁹ report, ubiquitous, perfect synthetic media means "distorting democratic discourse; manipulating elections; eroding trust in institutions; weakening journalism; exacerbating social divisions; undermining public safety; and inflicting hard-to-repair damage on the reputation of prominent individuals, including elected officials and candidates for office."

Location: 3,374

What if a large majority of white-collar tasks can be performed more efficiently by AI? In few areas will humans still be "better" than machines. I have long argued this is the more likely scenario.

Location: 3,377

These tools will only temporarily augment human intelligence. They will make us smarter and more efficient for a time, and will unlock enormous amounts of economic growth, but they are fundamentally labor replacing. They will eventually do cognitive labor more efficiently and more cheaply than many people working in administration, data entry, customer service (including making and receiving phone calls), writing emails, drafting summaries, translating documents, creating content, copywriting, and so on. In the face of an abundance of ultra-low-cost equivalents, the days of this kind of "cognitive manual labor" are numbered.

Location: 3,399

automation is unequivocally another fragility amplifier.

Location: 3,439

The full amplification of fragility is missed because it often appears as if these impacts were happening incrementally and in convenient silos. They are not. They stem from a single coherent and interrelated phenomenon manifesting itself in different ways. The reality is much more enmeshed, entwined, emergent, and chaotic than any sequential presentation can convey. Fragility, amplified. The nation-state, weakened.

Location: 3,445

The fully omni-use nature of the coming wave means it is found at every level, in every sector, every business, or subculture, or group, or bureaucracy, in every corner of our world. It produces trillions of dollars in new economic value while also destroying certain existing sources of wealth. Some individuals are greatly enabled; others stand to lose everything. Militarily it empowers some nation-states and militias alike. This is not, then, confined to amplifying specific points of fragility; it is, in the slightly longer term, about a transformation of the very ground on which society is built. And in this great redistribution of power, the state, already fragile and growing more so, is shaken to its core, its grand bargain left tattered and precarious.

Location: 3,478

Exponential technologies amplify everyone and everything. And that creates seemingly contradictory trends. Power is both concentrated and dispersed. Incumbents are both strengthened and weakened. Nation-states are both more fragile and at greater risk of slipping into abuses of unchecked power.

Location: 3,507

We are not quite heading for a neocolonial East India Company 2.0. But I do think we have to confront the sheer scale and influence that some boardrooms have not just over the subtle nudges and choice architectures that shape culture and politics today but, more importantly, over where this could lead in decades to come. They are empires of a sort, and with the coming wave their scale, influence, and capability are set to radically expand.

Location: 3,768

What does the social contract look like if a select group of “post-humans” engineer themselves to some unreachable intellectual or physical plane? How would this intersect with the dynamic of fragmenting politics, some enclaves trying to leave the whole behind?

Location: 3,786

The techno-libertarian movement takes Ronald Reagan’s 1981 dictum “Government is the problem” to its logical extreme, seeing government’s many flaws but not its immense benefits, believing that its regulatory and tax functions are destructive rate limiters with few upsides—for them at least. I find it deeply depressing that some of the most powerful and privileged take such a narrow and destructive view, but it adds a further impetus to fragmentation. This is a world where billionaires and latter-day prophets can build and run microstates; where non-state actors from corporations to communes to algorithms begin to overshadow the state from above but also from below.

Location: 3,852

On the one hand, societies could turn toward the kind of tech-enabled total surveillance we saw in the last chapter, a gut response enforcing hard mechanisms against wayward or uncontrolled technology. Security—at the price of freedom. Or humanity might step away from the technological frontier altogether. Although unlikely, it’s no answer. The only entity in principle capable of navigating this existential bind is the same system of nation-states currently falling apart, dragged down by the very forces it needs to contain. Over time, then, the implications of these technologies will push humanity to navigate a path between the poles of catastrophe and dystopia. This is the essential dilemma of our age.

Location: 3,941

Once technology reaches a critical capability, it isn’t enough for early pioneers to just build it

safely, as challenging as that undoubtedly is. Rather, true safety requires maintaining those standards across every single instance: a mammoth expectation given how fast and widely these are already diffusing.

Location: 4,015

A cataclysm would galvanize calls for an extreme surveillance apparatus to stop future such events. If or when something goes wrong with technology, how long before the crackdown starts? How could anyone plausibly argue against it in the face of a disaster? How long before the surveillance dystopia puts down roots, one creeping tendril at a time, and grows? As smaller-scale technology failures mount, calls for control increase. As control increases, checks and balances get whittled down, the ground shifts and makes way for further interventions, and a steady downward spiral to techno-dystopia begins.

Location: 4,030

If zombielike states will sleepwalk into catastrophe, their openness and growing chaos a petri dish for uncontained technology, authoritarian states are already gladly charging into just this techno-dystopia, setting the stage, technologically if not morally, for massive invasions of privacy and curtailments of liberty. And on the continuum¹⁰ between the two there is also a chance of the worst of all worlds: scattered but repressive surveillance and control apparatuses that still don't add up to a watertight system. Catastrophe and dystopia.

Location: 4,065

Our entire edifice is premised on the idea of long-term economic growth. And long-term economic growth is ultimately premised on the introduction and diffusion of new technologies. Whether it's the expectation of consuming more for less or getting ever more public service without paying more tax, or the idea that we can unsustainably degrade the environment while life keeps getting better indefinitely, the bargain—arguably the grand bargain itself—needs technology.

Location: 4,113

In 1955, toward the end of his life,²⁰ the mathematician John von Neumann wrote an essay called “Can We Survive Technology?” Foreshadowing the argument here, he believed that global society was “in a rapidly maturing crisis—a crisis attributable to the fact that the environment in which technological progress must occur has become both undersized and underorganized.” At the end of the essay, von Neumann puts survival as only “a possibility,” as well he might in the shadow of the mushroom cloud his own computer had made a reality. “For progress there is no cure,” he writes. “Any attempt to find automatically safe channels for the present explosive variety of progress must lead to frustration.”

Location: 4,133

Technology is the best and worst of us. There isn’t a neat one-sided approach that does it justice. The only coherent approach to technology is to see both sides at the same time. Over the last decade or so this dilemma has become even more pronounced, the task of tackling it more urgent. Look at the world and it seems that containment is not possible. Follow the consequences and something else becomes equally stark: for everyone’s sake, containment must be possible.

Location: 4,157

Saying “Regulation!” in the face of awesome technological change is the easy part. It’s also the classic pessimism-averse answer. It’s a simple way to shrug off the problem. On paper regulation looks enticing, even obvious and straightforward; suggesting it lets people sound smart, concerned, and even relieved. The unspoken implication being that it’s solvable, but it’s someone else’s problem. Look deeper, though, and the fissures become evident.

Location: 4,166

While garage amateurs gain access to more powerful tools and tech companies spend billions on R&D, most politicians are trapped in a twenty-four-hour news cycle of sound bites and photo ops. When a government has devolved to the point of simply lurching from crisis to crisis, it has little breathing room for tackling tectonic forces requiring deep domain expertise and careful judgment on uncertain timescales. It’s easier to ignore these issues in favor of low-hanging fruit more likely to win votes in the next election.

Location: 4,183

It’s not enough to have dozens of separate conversations about algorithmic bias or bio-risk or drone warfare or the economic impact of robotics or the privacy implications of quantum computing. It completely underplays how interrelated both causes and effects are. We need an approach that unifies these disparate conversations, encapsulating all those different dimensions of risk, a general-purpose concept for this general-purpose revolution.

Location: 4,192

The central problem for humanity in the twenty-first century is how we can nurture sufficient legitimate political power and wisdom, adequate technical mastery, and robust norms to constrain technologies to ensure they continue to do far more good than harm. How, in other words, we can contain the seemingly uncontainable.

Location: 4,204

novel threats are just exceptionally difficult for any government to navigate. That's not a flaw with the idea of government; it's an assessment of the scale of the challenge before us. When they are faced with something like an ACI that can pass my version of the Modern Turing Test, the response of even the most thoughtful, farsighted bureaucracies will resemble the response to COVID. Governments fight the last war, the last pandemic, regulate the last wave. Regulators regulate for things they can anticipate. This, meanwhile, is an age of surprises.

Location: 4,222

Most regulation walks a tightrope of competing interests. But in few areas other than frontier technology must it tackle something so widely diffused, so critical to the economy, and yet so fast evolving. All the noise and confusion makes clear how hard and complex any form of regulation is, especially amid accelerating change, and how, because of that, it will almost certainly leave gaps, falling short of effective containment.

Location: 4,230

Complex regulations refined over decades made roads and vehicles incrementally safer and more ordered, enabling their growth and spread. And yet 1.35 million people⁸ a year still die in traffic accidents. Regulation may lessen the negative effects, but it can't erase bad outcomes like crashes, pollution, or sprawl. We have decided that this is an acceptable human cost, given the benefits. That "we" is crucial. Regulation doesn't just rely on the passing of a new law. It is also about norms, structures of ownership, unwritten codes of compliance and honesty, arbitration procedures, contract enforcement, oversight mechanisms. All of this needs to be integrated and the public needs to buy in. This takes time—time we don't have.

Location: 4,241

ABOVE THE CUT AND thrust of legislative debate, nations are also caught in a contradiction. On the one hand, they are in a strategic competition to accelerate the development of technologies like AI and synthetic biology. Every nation wants to be, and be seen, at the technological frontier. It's a measure of national pride, of national security, and an existential imperative. On the other hand, they're desperate to regulate and manage these technologies—to contain them, not least for fear they will threaten the nation-state as the ultimate seat of power. The scary thing is that this assumes a best-case scenario of strong, reasonably competent, cohesive (liberal democratic) nation-states capable of working coherently as units internally and coordinating well internationally.

Location: 4,282

Contained technology is technology whose modes of failure are known, managed, and mitigated, a situation where the means to shape and govern technology escalate in parallel with its capabilities.

Location: 4,289

Containment will need to respond to the nature of a technology, and channel it in directions that are easier to control. Recall the four features of the coming wave: asymmetry, hyper-evolution, omni-use, and autonomy. Each feature must be viewed through the lens of containability.

Location: 4,373

Ten Steps Toward Containment

Location: 4,381

1. SAFETY: AN APOLLO PROGRAM FOR TECHNICAL SAFETY

Location: 4,398

Technical safety, up close, in the code, in the lab, is the first item on any containment agenda.

Location: 4,407

Physical segregation is just one aspect of transforming technical safety architecture to meet the challenge of the next wave.

Location: 4,412

Bodies like the Institute of Electrical and Electronics Engineers maintain more than two thousand technical safety standards on technologies ranging from autonomous robot development to machine learning. Biotech and pharma have operated under safety standards far beyond those of most software businesses for decades. It's worth remembering just how safe years of effort have made many existing technologies—and building on it.

Location: 4,415

Frontier AI safety research is still an undeveloped, nascent field focusing on keeping ever more autonomous systems from superseding our ability to understand or control them. I see these questions around control or value alignment as subsets of the wider containment problem. While billions are plowed into robotics, biotech, and AI, comparatively tiny amounts get spent on a technical safety framework equal to keeping them functionally contained.

Location: 4,425

Compared with the magnitude of what could go wrong, safety and ethics research on AI is marginal. Only a handful of institutions, owing to the challenges of resources, take technical safety issues seriously.

Location: 4,428

It's time for an Apollo program on AI safety and biosafety.

Location: 4,429

Concretely, a good proposal for legislation would be to require that a fixed portion—say, a minimum of 20 percent—of frontier corporate research and development budgets should be directed toward safety efforts, with an obligation to publish material findings to a government working group so that progress can be tracked and shared.

Location: 4,453

Explanation is another huge technical safety frontier. Recall that at present no one can explain why, precisely, a model produces the outputs it does. Devising ways for models to comprehensively explain their decisions or open them to scrutiny has become a critical technical puzzle for safety researchers.

Location: 4,462

The computer scientist Stuart Russell⁸ proposes using the kind of built-in systematic doubt we are exploring at Inflection to create what he calls “provably beneficial AI.” Rather than give an AI a set of fixed external objectives contained in what’s known as a written constitution, he recommends that systems gingerly infer our preferences and ends.

Location: 4,474

The highest-level challenge, whether in synthetic biology, robotics, or AI, is building a bulletproof off switch, a means of closing down any technology threatening to run out of control.

Location: 4,485

2. AUDITS: KNOWLEDGE IS POWER; POWER IS CONTROL

Location: 4,490

Trust comes from transparency. We absolutely need to be able to verify, at every level, the safety, integrity, or uncompromised nature of a system. That in turn is about access rights and audit capacity, about adversarially testing systems, having teams of white hat hackers or even AIs probing weaknesses, flaws, and biases. It’s about building technology in an entirely different way, with tools and techniques that don’t exist yet.

Location: 4,498

A few years ago I co-founded a cross-industry and civil society organization called the Partnership on AI to help with this kind of work. We launched it with the support of all the major technology companies, including DeepMind, Google, Facebook, Apple, Microsoft, IBM, and OpenAI, along with scores of expert civil society groups, including the ACLU, the EFF, Oxfam, UNDP, and twenty others. Shortly after, it kick-started an AI Incidents Database, designed for confidentially reporting on safety events to share lessons with other developers. It has now collected more than twelve hundred reports.

Location: 4,511

It’s also time to create government-funded red teams that would rigorously attack and stress test every system, ensuring that insights discovered along the way are shared widely across the industry.

Location: 4,517

APIs that let others use foundational AI services should not be blindly open, but rather come with “know your customer” checks, as with, say, portions of the banking industry.

Location: 4,529

Screening all DNA synthesis would be a major bio-risk reduction exercise and would not, in my view, unduly curb civil liberties.

Location: 4,531

Pre-vetting DNA synthesis or data inputs to AI models would front-load audits before systems were deployed, reducing risk.

Location: 4,548

3. CHOKE POINTS: BUY TIME

Location: 4,583

Right now technology is driven by the power of incentives rather than the pace of containment. Export controls like the United States’ semiconductor gambit have all kinds of uncertain implications for great power competition, arms races, and the future, but almost everyone agrees on one thing: this will slow down at least some technological development in China, and by extension the world.

Location: 4,589

In AI, the lion’s share of the most advanced GPUs essential to the latest models are designed by one company, the American firm NVIDIA. Most of its chips are manufactured by one company, TSMC, in Taiwan, the most advanced in just a single building, the world’s most sophisticated and expensive factory. TSMC’s machinery to make these chips comes from a single supplier, the Dutch firm ASML, by far Europe’s most valuable and important tech company. ASML’s machines,¹⁸ which use a technique known as extreme ultraviolet lithography and produce chips at levels of astonishing atomic precision, are among the most complex manufactured goods in history. These three companies have¹⁹ a choke hold on cutting-edge chips, a technology so physically constrained that one estimate argues they cost up to \$10 billion per kilogram.

Location: 4,604

So, as negative impacts become clear, we must use these choke points to create sensible rate-limiting factors, checks on the speed of development, to better ensure that good sense is implemented as fast as the science evolves.

Location: 4,610

4. MAKERS: CRITICS SHOULD BUILD IT

Note | Location: 4,610

Location: 4,623

In their own way, tech's critics fall into a form of the pessimism-aversion trap that is hardwired into techno/political/business elites. Many who ridicule overly optimistic technologists stick to writing theoretical oversight frameworks or op-eds calling for regulation. If you believe technology is important and powerful, and you follow the implications of these critiques, such responses are clearly inadequate. Even the critics duck the true reality in front of them. Indeed, at times shrill criticism²² just becomes part of the same hype cycle as technology itself.

Location: 4,628

Credible critics must be practitioners.

Location: 4,651

Major shortfalls²⁴ in bringing nontechnical perspectives and diverse voices into the discussion are still all too commonplace, however: contained technology is a project requiring all kinds of disciplines and perspectives. Proactively hiring to that effect is a must.

Location: 4,655

BUSINESSES: PROFIT + PURPOSE

Location: 4,670

After our acquisition by Google, my co-founders and I spent years trying to build an ethics charter into the legal fabric of the company, endlessly arguing about how much of this charter could be public, how much of DeepMind's work could be further subject to independent oversight and scrutiny. Our goal in these discussions was always to ensure that unprecedented technology was matched by unprecedented governance. Our proposal was to spin DeepMind out as a new form of "global interest company," with a fully independent board of trustees separate from and in addition to the board of directors tasked with operationally running the company. Membership, decision-making, and even some of the board's reasoning would be more public. Transparency, accountability, ethics—these would be not just corporate PR but foundational, legally binding, and built into everything the company did. We felt this would let us work in an open way, proactively learning how companies could be resilient and modern long-term stewards of exponential technologies.

Location: 4,684

In hindsight it was just too much for Google at the time. Lawyers were retained, years of intense negotiations took place, but there didn't seem to be a way to square the circle. In the end we couldn't find an answer that would satisfy everyone. DeepMind continued as a normal unit within Google with no formal legal independence, operating just as a separate brand. It was a foundational lesson for me: shareholder capitalism works because it is simple and clear, and governance models too have a tendency to default to the simple and clear.

Location: 4,713

Despite the awesome scale of the rewards on offer, entrepreneurs, execs, and employees alike should keep pushing and exploring corporate forms that can better accommodate the challenge of containment.

Location: 4,724

GOVERNMENTS: SURVIVE, REFORM, REGULATE

Location: 4,731

I think the government needs to get way more involved, back to building real technology, setting standards, and nurturing in-house capability. It needs to compete for talent and hardware in the open market. There's no two ways about it: this is expensive and will come with wasteful mistakes. But proactive governments will exert far greater control than if they just commission services and live off outsourced expertise and tech owned and operated elsewhere.

Location: 4,734

Accountability is enabled by deep understanding. Ownership gives control. Both require governments to get their hands dirty. Although today companies have²⁷ taken the lead, much of the most speculative fundamental research is still funded by governments. U.S. federal government expenditure on R&D is at an all-time-low share of the total—just 20 percent—but still amounts to a not inconsiderable \$179 billion per year.

Location: 4,742

Governments should not rely on management consultants, contractors, or other third-party suppliers. Full-time, well-respected staffers who are properly compensated, competitively with the private sector, should be a core part of the solution. Instead, private sector salaries²⁹ can be ten times their public sector equivalents in national critical roles: it's unsustainable.

Location: 4,756

Regulation alone doesn't get us to containment, but any discussion that doesn't involve regulation is doomed. Regulation should focus on those incentives, better aligning individuals, states, companies, and the public as a whole with safety and security while building in the possibility of a hard brake. Certain use cases, like AI for electioneering, should be prohibited by law as part of the package.

Location: 4,774

The most sophisticated AI systems or synthesizers or quantum computers should be produced only by responsible certified developers. As part of their license, they would need to subscribe to clear, binding security and safety standards, following rules, running risk assessments, keeping records, closely monitoring live deployments. Just as you cannot simply launch a rocket into space without FAA approval, so tomorrow you shouldn't simply be able to release a state-of-the-art AI.

Location: 4,808

ALLIANCES: TIME FOR TREATIES

Location: 4,842

A golden age of techno-diplomacy needs to emerge from the era of arms races.

Location: 4,858

The SecureDNA project is a good example, laying out a path for governing synthetic biology similarly to how chemical weapons have been curtailed. If China and the United States could create, say, a shared bio-risk observatory, encompassing everything from advanced R&D to deployed commercial applications, it would be a precious area of collaboration to build on.

Location: 4,865

As the century wears on, the lesson of the Cold War will have to be relearned: there is no path to technological safety without working with your adversaries.

Location: 4,874

A dedicated regulator that navigates contentious geo-politics (as much as possible), avoids overreach, and performs a pragmatic monitoring function on broadly objective criteria is urgently needed. Think of something like the International Atomic Energy Agency or even a trade body like the International Air Transport Association.

Location: 4,884

CULTURE: RESPECTFULLY EMBRACING FAILURE

Location: 4,905

Fear of failure and public opprobrium is leading to stasis. Immediate self-reporting of problems should be a baseline for individuals and organizations alike. But rather than being commended for experimentation, companies and teams are hung out to dry. Doing the right thing only triggers a backlash of cynicism, Twitter flaming, and vicious public point scoring. Why would anyone actually admit their mistakes in this context? This has got to stop if we want to produce better, more responsible, more containable technologies.

Location: 4,920

In AI, capabilities like recursive self-improvement and autonomy are, I think, boundaries we should not cross. This will have technical and legal components, but also needs moral, emotional, cultural buy-in from the people and organizations closest to it.

Location: 4,946

MOVEMENTS: PEOPLE POWER

Location: 4,977

Change happens when people demand it. The “we” that builds technology is scattered, subject to a mass of competing and different national, commercial, and research incentives. The more the “we” that is subject to it speaks clearly in one voice, a critical public mass agitating for change, demanding an alignment of approaches, the better chance of good outcomes.

Location: 4,981

THE NARROW PATH: THE ONLY WAY IS THROUGH

Location: 5,011

Put all the elements here together and there is an outline of what will meet and match the coming wave. 1. Technical safety Concrete technical measures to alleviate possible harms and maintain control. 2. Audits A means of ensuring the transparency and accountability of technology. 3. Choke points Levers to slow development and buy time for regulators and defensive technologies. 4. Makers Ensuring responsible developers build appropriate controls into technology from the start. 5. Businesses Aligning the incentives of the organizations behind technology with its containment. 6. Government Supporting governments, allowing them to build technology, regulate technology, and implement mitigation measures. 7. Alliances Creating a system of international cooperation to harmonize laws and programs. 8. Culture A culture of sharing learning and failures to quickly disseminate means of addressing them. 9. Movements All of this needs public input at every level, including to put pressure on each component and make it accountable. Step 10 is about coherence, ensuring that each element works in harmony with the others, that containment is a virtuous circle of mutually reinforcing measures and not a gap-filled cacophony of competing programs.

Note | Location: 5,024

Location: 5,043

Safe, contained technology is, like liberal democracy, not a final end state; rather, it is an ongoing process, a delicate equilibrium that must be actively maintained, constantly fought for and protected.

Location: 5,050

Open-source has been a boon to technological development and a major spur to progress more widely. But it’s not an appropriate philosophy for powerful AI models or synthetic organisms; here it should be banned. They should not be shared, let alone deployed or developed, without rigorous due process.

Location: 5,060

Some measure of anti-proliferation is necessary. And yes, let’s not shy away from the facts; that means real censorship, possibly well beyond national borders.

Location: 5,156

Too many visions of the future start with what technology can or might do and work from there. That's completely the wrong foundation. Technologists should focus not just on the engineering minutiae but on helping to imagine and realize a richer, social, human future in the broadest sense, a complex tapestry of which technology is just one strand. Technology is central to how the future will unfold—that's undoubtedly true—but technology is not the point of the future, or what's really at stake. We are.